

| Question |     |      |    | Marking details                                                                                                                                                                                                                                                                                                                                                                                             | Marks available |     |     |       |       |      |
|----------|-----|------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) | (i)  |    | The highest levels (of NO <sub>2</sub> ) are {along major roads / where there are most vehicles / owtte}<br>Accept specifically named roads                                                                                                                                                                                                                                                                 |                 |     | 1   | 1     |       |      |
|          |     | (ii) | I  | circle on black rectangle dot on M4 near Port Talbot                                                                                                                                                                                                                                                                                                                                                        |                 |     | 1   | 1     |       |      |
|          |     |      | II | the level (of NO <sub>2</sub> most) exceeds WHO limits / values >60 compared with {WHO limit / 40}                                                                                                                                                                                                                                                                                                          |                 |     | 1   | 1     |       |      |
|          | (b) | (i)  |    | <ul style="list-style-type: none"> <li>Critical value is 1.701 (at p=0.05) (1)</li> <li>Because t value is greater than {critical value/ 1.701} / 1.769 &gt; 1.701 / ORA (1)</li> <li>Reject null hypothesis (1)</li> <li>There is a <u>significant</u> difference between the <u>means</u> / the difference between the <u>means</u> is not due to chance (1)</li> </ul> Ecf from incorrect critical value |                 | 3   | 1   | 4     | 4     |      |
|          |     | (ii) |    | Emissions were (significantly) higher after closure / ORA (1)<br>Keeping junction open will result in {lower NO <sub>2</sub> levels / better air quality} (1)<br><b>Ecf from (i)</b><br>There was no significant difference in (mean) emissions (1)<br>Closing the junction will not result in any changes in the NO <sub>2</sub> levels / air quality} (1)                                                 |                 |     | 2   | 2     |       |      |